

# Notes: Albertini, Poirier, and Terriau (RES, 2026)

## The Impact of EITC on Education, Labour Market Trajectories, and Inequalities

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# 1 Big picture

- **Question.** The paper asks whether the Earned Income Tax Credit (EITC), a policy designed to reward work among low-income workers, has long-run consequences for schooling, labour market trajectories, and inequality.
- **Core tension.** The EITC raises the return to working in low-paid jobs. That can increase participation today, but it can also reduce the relative payoff to education because the policy makes low-skill employment more attractive.
- **Empirical first step.** The authors exploit discontinuities in state EITC generosity at US state borders. They compare contiguous Public Use Microdata Area (PUMA) pairs on opposite sides of state borders using ACS data from 2005–2018.
- **Main empirical result.** A one-percentage-point increase in the state EITC, expressed as a percent of the federal EITC, increases the high-school dropout rate by about 0.07 percentage point. The estimate is robust to controls for compulsory schooling, minimum wages, demographics, parental environment, local labour markets, and other state policies.
- **Structural contribution.** The paper builds a life-cycle directed-search model in which schooling, human capital, job search, vacancies, hirings, separations, hours, and transfers are endogenous. This lets the authors study both immediate labour-supply responses and long-run education/composition effects.
- **Mechanism.** An EITC targeted at low-wage workers increases the payoff to low-skill jobs. This encourages some youth to stop education earlier, increases the long-run share of low-skilled workers, and lowers the economy’s human-capital composition.
- **Hours channel.** Many recipients are in the phase-out range or outside the EITC schedule. Increasing EITC generosity can induce them to reduce hours in order to qualify for the credit or obtain a higher credit.
- **Long-run outcome.** The policy increases participation for low-paid workers but reduces education, aggregate hours, employment, and lifetime earnings enough that it does not significantly reduce income inequality in the long run.
- **Optimal policy lesson.** The problem is not the idea of in-work benefits per se, but the shape of the EITC. A more welfare-improving schedule has lower maximum benefits, broader phase-in and phase-out ranges, reaches higher incomes, and can perform better when combined with a minimum-hours eligibility requirement.
- **Takeaway.** Evaluating in-work benefits only by short-run employment responses misses an important Lucas-critique channel: people adjust education before they enter the labour market.

## 2 Data, measurement, and identification details

### 2.1 The treatment variable

The key treatment is the state EITC rate, measured as a percentage of the federal EITC. If a state offers a 10 percent state EITC, then an individual who qualifies for a federal credit receives an additional credit equal to 10 percent of the federal amount.

- This treatment varies by state and year.
- The federal EITC schedule varies by family status and earned income, but the empirical treatment is the state supplement rate.
- The state supplement creates a clean geographic discontinuity at state borders.

## **2.2 The outcome variables**

- Main outcome: high-school dropout rate for 18- to 24-year-olds.
- Secondary education outcomes: high-school completion and college enrolment.
- Additional labour-market outcomes: employment, hours, earnings, and hourly wages.

## **2.3 Why border-PUMA comparisons are credible**

- State policy changes at state borders, but many local economic conditions are shared across neighbouring PUMAs.
- PUMA fixed effects remove time-invariant local differences.
- Pair-time fixed effects remove border-pair shocks in a given period.
- Controls absorb demographic, parental, labour-market, and other state-policy changes.

## **2.4 Main empirical limitations**

- State EITC adoption is not randomized.
- Border PUMAs may still differ in unobserved ways, though pair-time fixed effects reduce this concern.
- The reduced-form estimate captures an education response to state EITC variation, but the full long-run effect requires a structural model.
- Migration and commuting spillovers are possible; the paper addresses them with alternative interior control PUMAs.

### 3 Model mechanics and key equations

#### 3.1 EITC as a non-linear tax-transfer schedule

The EITC creates four regimes:

1. phase-in: benefits rise with earned income;
2. plateau: benefits equal the maximum credit;
3. phase-out: benefits decline as earnings rise;
4. no-credit range: benefits are zero.

**Interpretation:** The same increase in EITC generosity can raise hours in the phase-in range, lower hours in the phase-out range, and change education before labour-market entry.

#### 3.2 Two-sided labour-market decisions

A job match survives only if both sides want to continue:

$$\text{worker continues if } W(h, a, f, x, e) > U(h, a, f, e), \quad (1)$$

$$\text{firm continues if } J(h, a, f, x, e) > V(h, a, f, e). \quad (2)$$

**Interpretation:** EITC policy affects both the worker's outside option and the firm's value of a match through hours and productivity. Hence it can change job creation and destruction, not just labour supply.

#### 3.3 Education choice

Before labour-market entry, individuals choose education by comparing expected values net of direct and opportunity costs. Higher education raises expected wages and job prospects, but it has private and public costs and delays labour-market entry.

**Interpretation:** The EITC changes the expected value of each education path. If low-skill work is subsidized more heavily, the relative gain from completing high school or college can fall.

#### 3.4 Why transition dynamics take decades

- The education decision is made by new cohorts before labour-market entry.
- Existing workers cannot retroactively change their education.
- Therefore, the skill composition changes gradually as new cohorts enter and older cohorts retire.
- The paper estimates that reaching the new steady state can take roughly half a century.

**Interpretation:** This is why short-run evaluations of the EITC can miss the central long-run effect. The policy's full consequences appear only after the education composition of the labour force changes.

### 3.3. Productivity and income

Given the amount of heterogeneity in the model, it is useful to define the productivity and wage corresponding to a firm-worker pair. Productivity is given by:

$$y(h, x, \ell, e) = A(e)hx\ell, \quad (2)$$

where  $A(e)$  is an education-dependent scaling parameter and  $\ell \in [0, 1]$  corresponds to the number of working hours supplied by the individual.

Following [Burdett et al. \(2011\)](#), [Bagger et al. \(2014\)](#), and [Blundell et al. \(2016\)](#), we consider pre-tax hourly wages to be defined by the following rule:

$$w(h, x, e) = \max(w_{\min}^h, A(e)xh^{\bar{\alpha}(e)}), \quad (3)$$

where  $w_{\min}^h$  is the hourly minimum wage and  $\bar{\alpha}(e)$  is the elasticity of wages with respect to human capital.

As in [Blundell et al. \(2016\)](#), we compute the net social transfer (welfare less taxes) depending on age, family status, education, and labour force status. We map these revenues into the model's state space by defining  $b(a, f, e, j)$  as the net monetary gain derived from all tax and welfare programmes.  $j = \{u, n\}$  denotes job status, *i.e.* unemployed (*u*) or employed (*n*).

### 3.4. Modelling of the EITC

Our model seeks to capture the major characteristics of the EITC. For simplicity, we denote earned income by  $\tilde{w} = w\ell$ . Workers can receive the tax credit depending on their earned income  $\tilde{w}$ .<sup>16</sup> Let  $\bar{\tau}$  denote the basic social security contribution rate, and let  $c_{\max}$  denote the maximum tax credit.  $w_m$  and  $\bar{w}$  are, respectively, the minimum and maximum earned income to benefit from the maximum tax credit (the lower and upper bounds of the plateau range).  $\bar{w}$  corresponds to the maximum earned income to benefit from the EITC (the upper bound of the phase-out range).  $\tau(\tilde{w}, f)$  denotes the social security contribution net of the tax credit. Formally, it may be written:

$$\tau(\tilde{w}, f) = \underbrace{\bar{\tau}\tilde{w}}_{\text{basic tax}} - \underbrace{\max\left(\min\left(\frac{\tilde{w}}{w_m(f)}, 1, \frac{\bar{w}(f) - \tilde{w}}{\bar{w}(f) - \underline{w}(f)}\right), 0\right)c_{\max}(f)}_{\text{EITC}}. \quad (4)$$

The parameters depend on family status, as in the original rule presented in Figure 1. Note that:

(a) Original equations for productivity, wages, and EITC schedule

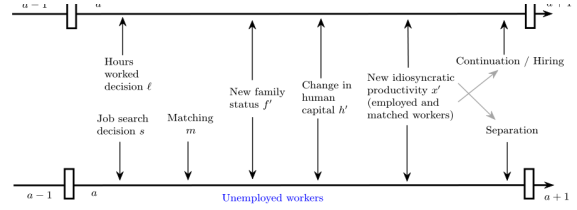


FIGURE 7  
Timing of events

(b) Original timing of events

### 3.7. Bellman equations

*Optimal continuation and acceptance decisions.*  $W$ ,  $U$ ,  $J$ , and  $V$  denote the respective value functions for an employed worker, an unemployed worker, a filled job, and a vacant job. The optimal decisions are given by:

$$\Omega(h, a, f, x, e) = \mathbb{1}^F(h, a, f, x, e) \max[W(h, a, f, x, e), U(h, a, f, e)] + (1 - \mathbb{1}^F(h, a, f, x, e))U(h, a, f, e), \quad (8)$$

$$\Lambda(h, a, f, x, e) = \mathbb{1}^W(h, a, f, x, e) \max[J(h, a, f, x, e), V(h, a, f, e)] + (1 - \mathbb{1}^W(h, a, f, x, e))V(h, a, f, e), \quad (9)$$

with,

$$\mathbb{1}^F(h, a, f, x, e) = \begin{cases} 1 & \text{if } J(h, a, f, x, e) > V(h, a, f, e) \\ 0 & \text{otherwise} \end{cases}, \quad (10)$$

$$\mathbb{1}^W(h, a, f, x, e) = \begin{cases} 1 & \text{if } W(h, a, f, x, e) > U(h, a, f, e) \\ 0 & \text{otherwise} \end{cases}. \quad (11)$$

For an employed worker,  $\Omega$  represents the optimal decision to continue the match. If the value of unemployment exceeds that of employment, the employee will quit. Otherwise, the relationship continues. For an unemployed worker,  $\Omega$  represents the decision to accept or turn down the job offer.  $\Omega$  depends on whether the firm is willing to continue the relationship,  $\mathbb{1}^F(h, a, f, x, e) = 1$ , or not. If not,  $\mathbb{1}^F(h, a, f, x, e) = 0$ , the firm unilaterally decides to end the relationship. We thus have inefficient separations (layoffs). Similarly, for a firm with a filled job,  $\Lambda$  provides the decision to maintain the employment relationship or lay off the worker. For a firm with an unfilled position,  $\Lambda$  determines whether a vacant job is transformed into a filled job.  $\Lambda$  depends on whether the employed worker is willing to continue the relationship or if the unemployed worker accepts the job offer, that is  $\mathbb{1}^W(h, a, f, x, e) = 1$ . Otherwise, the worker unilaterally decides to end the relationship (quit and job refusal).

*Employed workers.* We assume that employed workers choose how many hours they are willing to work, given the statutory constraint imposed by the EITC. The value function for an employed worker is expressed as follows:

$$W(h, a, f, x, e) = \max_{\ell \in [0, 1]} \left\{ w(h, x, e)\ell - \tau(w(h, x, e)\ell) + b(a, f, e, n) - \eta(a, e)^{\frac{1+\phi}{1-\phi}} \right\} + \beta \sum_{h'} \mu_n(h, h') \int \int \left[ (1 - \delta_e)\Omega(h', a', f', x', e) + \delta_e U(h', a', f', e) \right] dX(x'|x) dF_{a,e}(f'|f). \quad (12)$$

An employed worker<sup>17</sup> earns  $w\ell$ , pays taxes net of the EITC  $\tau(w\ell)$ , and receives net social transfer  $b$ . The fourth term on the right-hand side corresponds to the disutility from working.

(b) Original worker/value-function excerpt

### 3.5. Matching

Following [Menzio et al. \(2016\)](#), we consider directed search over educational attainment  $e$ , age  $a$ , family status  $f$ , and human capital  $h$ . The number of hires per unit of time in each submarket is given by the following matching function:

$$m(h, a, f, e) = m(s(h, a, f, e)u(h, a, f, e), v(h, a, f, e)), \quad (5)$$

where  $u \geq 0$  is the mass of unemployed workers,  $v$  denotes the mass of vacancies, and  $s$  stands for endogenous search effort. The matching function (5) is increasing and concave in its two arguments and exhibits decreasing return to scale. The job-finding probability per efficiency unit of worker search  $p$  and the vacancy-filling probability  $q$  are defined as follows:

$$p(h, a, f, e) = m(h, a, f, e)/(u(h, a, f, e)s(h, a, f, e)), \quad (6)$$

$$q(h, a, f, e) = m(h, a, f, e)/v(h, a, f, e). \quad (7)$$

(a) Original matching function and probabilities

than more-educated workers to occupy jobs that are more physically demanding and detrimental to their health. Consequently, the disutility from working may increase with age and differ across educational attainment levels.  $\phi$  represents the inverse of Frisch elasticity. The first-order condition with respect to  $\ell$  provides the optimal decision on working hours:

$$w - w \frac{\partial \tau(w\ell)}{\partial \ell} - \eta(a, e)\ell^\phi = 0. \quad (13)$$

Given the nondifferentiability of the tax function, we solve the optimal working hours problem  $\ell$  numerically. The last term on the right-hand side shows the expected value of employment given the separation rate shock and changes in human capital, age, idiosyncratic productivity, and family status.

*Unemployed workers.* An unemployed worker receives an income  $b(a, f, e, u)$ . She decides her optimal search effort  $s$  given the search cost function  $k(s)$ , with  $k'(s) > 0$  and  $k''(s) \geq 0$ , and the expected gain from searching. The value function is expressed as follows:

$$U(h, a, f, e) = \max_s \left\{ \int \int \left[ \frac{b(a, f, e, u) - k(s) + \beta \sum_{h'} \mu_u(h, h') \times (1 - p(h, a, f, e)s)U(h', a', f', e)}{+ p(h, a, f, e)s\Omega(h', a', f', x', e)} \right] dX_0(x')dF_{a,e}(f'|f) \right\}. \quad (14)$$

An unemployed worker is matched with a vacant job at the rate  $p \times s$  and decides whether to accept or reject the offer. Note that as long as she remains unemployed, her human capital is likely to depreciate with probability  $\psi_u$ . The first-order condition with respect to  $s$  provides the optimal search effort decision:

$$k'(s) = p(h, a, f, e)\beta \sum_{h'} \mu_u(h, h') \int \int \Omega(h', a', f', x', e) - U(h', a', f', e)dX_0(x')dF_{a,e}(f'|f). \quad (15)$$

Original unemployed-worker value and search-effort excerpt

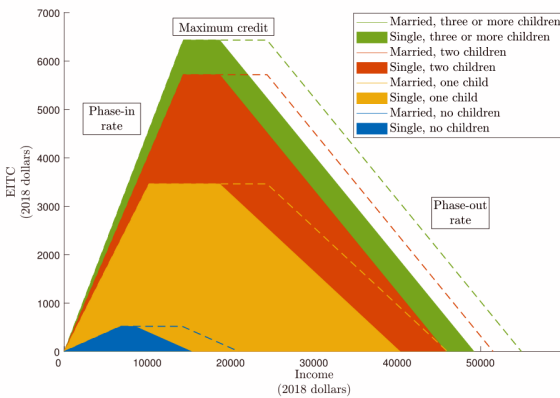
## 4 Tables and figures: read and economic message

### 4.1 Figures 1–3 and Table 1: EITC schedules and state supplements

Read:

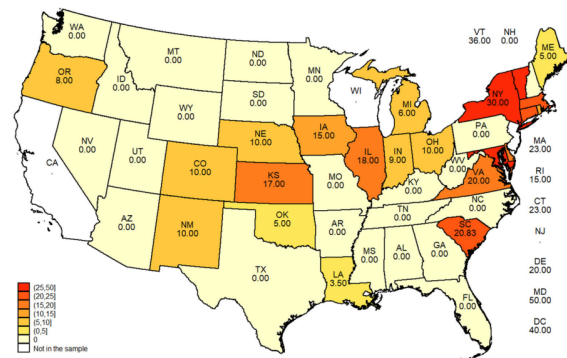
- Figure 1 shows the federal EITC’s trapezoidal schedule in 2018: phase-in, plateau, phase-out, and zero-credit regions.
- Figure 2 shows state-level heterogeneity in state EITC supplements as a percentage of the federal credit.
- Figure 3 and Table 1 translate state EITC rates into dollar gains for different household types. A 50 percent state supplement can be worth more than 3,000 dollars for a single person with three or more children in the extreme case shown.

**Economic message:** The EITC is large enough to affect incentives, and state supplements generate meaningful geographic policy variation.



Source: Internal Revenue Service

(a) Figure 1: Federal EITC schedule, 2018



(b) Figure 2: State EITC map, 2018

### 4.2 Figure 4, Table 2, and Figure 5: Border-PUMA design and summary statistics

Read:

- Figure 4 illustrates the contiguous-border PUMA comparison using New York and Pennsylvania.
- Table 2 shows that the border-pair sample has 401 PUMA pairs and an average state EITC of 7.54 percent of the federal EITC.
- Among 18- to 24-year-olds who have left high school in the border-pair sample, 8.66 percent have less than high school, 48.59 percent have completed high school, and 42.74 percent have more than high school.
- Figure 5 shows that the number of PUMA pairs with a state EITC differential increases over the sample period and that the average difference in state EITC is nontrivial.

**Economic message:** The empirical strategy has both geographic comparability and enough policy variation to estimate education effects.

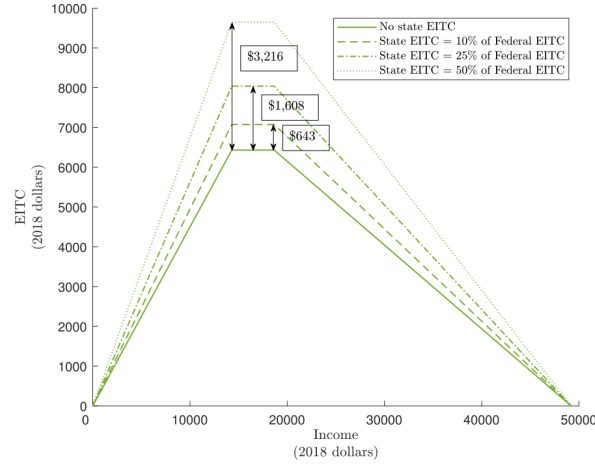


FIGURE 3  
State EITC (in dollars), 2018

(for a single person with three or more children)

TABLE 1  
State EITC, 2018 (for different types of households)

|                                 | State EITC<br>= 10% of Federal EITC |      |      |      | State EITC<br>= 25% of Federal EITC |      |       |       | State EITC<br>= 50% of Federal EITC |       |       |       |
|---------------------------------|-------------------------------------|------|------|------|-------------------------------------|------|-------|-------|-------------------------------------|-------|-------|-------|
|                                 | Number of children                  |      |      |      | Number of children                  |      |       |       | Number of children                  |       |       |       |
|                                 | 0                                   | 1    | 2    | 3    | 0                                   | 1    | 2     | 3     | 0                                   | 1     | 2     | 3     |
| Annual benefit (in dollars)     | 52                                  | 346  | 572  | 643  | 130                                 | 865  | 1,429 | 1,608 | 260                                 | 1,731 | 2,858 | 3,216 |
| Annual benefit (in % of income) | 0.77                                | 3.40 | 4.00 | 4.50 | 1.91                                | 8.50 | 10.00 | 11.25 | 3.83                                | 17.00 | 20.00 | 22.50 |

Notes: We consider here the extreme case of a single person receiving the maximum credit given the minimum income to be eligible. This represents the upper limit of earnings from the EITC (in % of income).

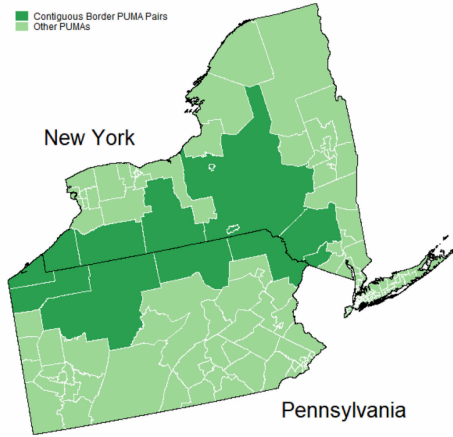


FIGURE 4

Contiguous-border PUMA pairs: Example for the states of New York and Pennsylvania

(a) Figure 4: contiguous-border PUMA pairs

TABLE 2  
Descriptive statistics

| VARIABLES                                            | (1)             |         | (2)                                |         |
|------------------------------------------------------|-----------------|---------|------------------------------------|---------|
|                                                      | All-PUMA sample |         | Contiguous border PUMA-Pair sample |         |
|                                                      | Mean            | s.d.    | Mean                               | s.d.    |
| <b>States</b>                                        |                 |         |                                    |         |
| State EITC                                           | 9.6015          | 13.7397 | 7.5412                             | 12.6949 |
| Compulsory school age                                | 16.7503         | 0.8572  | 16.8576                            | 0.8584  |
| Minimum wage                                         | 6.6297          | 2.4495  | 5.9824                             | 2.8283  |
| <b>18- to 24-year-olds who have left high school</b> |                 |         |                                    |         |
| Male                                                 | 0.5036          | 0.0549  | 0.5105                             | 0.0455  |
| White                                                | 0.7470          | 0.1818  | 0.7952                             | 0.1547  |
| Black                                                | 0.1300          | 0.1461  | 0.1127                             | 0.1366  |
| Employed                                             | 0.6056          | 0.1023  | 0.6023                             | 0.0889  |
| Less than high school                                | 7.6105          | 5.3272  | 8.6618                             | 4.6857  |
| High school                                          | 45.0566         | 9.2191  | 48.5941                            | 7.2249  |
| More than high school                                | 47.3329         | 11.7087 | 42.7440                            | 9.2444  |
| Number of PUMAs                                      | 897             |         | 314                                |         |
| Number of PUMA pairs                                 | –               |         | 401                                |         |
| Number of states                                     | 46              |         | 46                                 |         |

Notes: Sample means are reported for all PUMAs (1) and for all contiguous-border PUMA pairs (2).

The state EITC is expressed as a percentage of the federal EITC.

Source: ACS 1-year, 2005–18.

(b) Table 2: descriptive statistics

### 4.3 Table 3 and Figure 6: Main reduced-form result and spillover check

Read:

- Table 3 reports that the state EITC coefficient is about 0.071 in all four columns.
- The estimate is significant at the five percent level and robust to compulsory-schooling and minimum-wage controls.



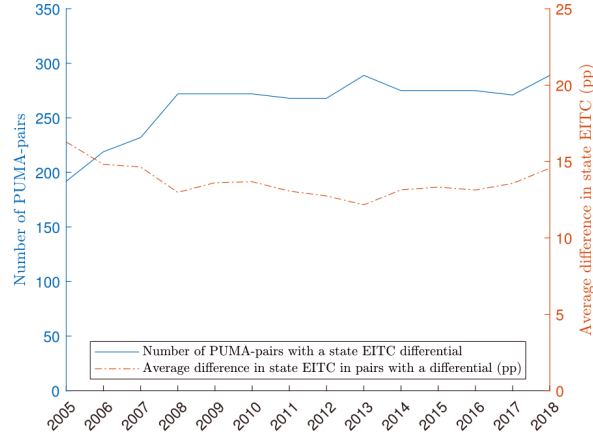


FIGURE 5  
Number of PUMA pairs with a state EITC differential, and average difference in state EITC in pairs with a differential  
Source: ACS 1-year, 2005–18

- Interpreted literally, a one-percentage-point increase in the state EITC raises the dropout rate by about 0.07 percentage point.
- Figure 6 explains the alternative control group used to test spillovers: the control can be a co-contiguous interior PUMA rather than an immediate border PUMA.

**Economic message:** The reduced-form evidence is the anchor for the model’s education channel. It shows that EITC generosity can move schooling choices.

| VARIABLES                  | (1)                  | (2)                  | (3)                  | (4)                  |
|----------------------------|----------------------|----------------------|----------------------|----------------------|
| State EITC                 | 0.0709**<br>(0.0360) | 0.0709**<br>(0.0358) | 0.0710**<br>(0.0358) | 0.0710**<br>(0.0358) |
| <b>Controls</b>            |                      |                      |                      |                      |
| PUMA pair × period dummies | YES                  | YES                  | YES                  | YES                  |
| Demographics               | YES                  | YES                  | YES                  | YES                  |
| Parental environment       | YES                  | YES                  | YES                  | YES                  |
| Labour market              | YES                  | YES                  | YES                  | YES                  |
| State policy               | YES                  | YES                  | YES                  | YES                  |
| <b>Additional controls</b> |                      |                      |                      |                      |
| Compulsory school age      | NO                   | YES                  | NO                   | YES                  |
| Minimum wage               | NO                   | NO                   | YES                  | YES                  |
| Number of periods          | 14                   | 14                   | 14                   | 14                   |
| Number of PUMA pairs       | 401                  | 401                  | 401                  | 401                  |
| R-squared                  | 0.8034               | 0.8034               | 0.8048               | 0.8048               |

Notes: Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

Source: ACS 1-year, 2005–18.

**Demographics:** sex, race; **Parental environment:** parents’ education, income, number of EITC-eligible children; **Labour market:** employment rate by education level; **State policy:** Supplemental Security Income, aid to families with dependent children, general assistance, food stamps.

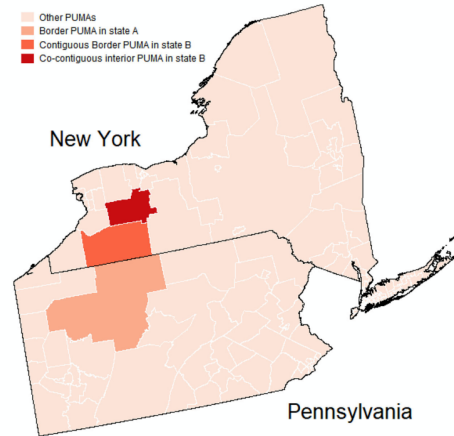


FIGURE 6  
Robustness test - Alternative control group to test for spillover effects  
Source: ACS 1-year

(a) Table 3: effect of EITC on high-school dropout rate

(b) Figure 6: alternative control group for spillovers

#### 4.4 Tables 4–6: Calibration, education targets, and estimated parameters

Read:

- Table 4 lists externally calibrated parameters, including ages, discount factor, human-capital range, matching-function elasticity, ability distribution, education costs, and EITC schedule parameters.
- Table 5 shows education targets. The benchmark distribution is approximately 9.48 percent low skill, 30.69 percent middle skill, and 59.83 percent high skill.

- Table 6 reports internally estimated structural parameters governing human capital, wages, matching, vacancy costs, separation, productivity, labour disutility, and education costs.

**Economic message:** The structural model is calibrated to reproduce both education shares and life-cycle labour-market moments. The reduced-form education elasticity becomes a discipline on the education-cost parameters.

TABLE 4  
Calibrated parameters

| Parameter                     | Symbol                 | Value               |
|-------------------------------|------------------------|---------------------|
| Initial age                   | $a_0$                  | 16                  |
| Final age                     | $a_A$                  | 67                  |
| Discount factor               | $\beta$                | 0.9966              |
| Human capital range           | $[h_1, h_H]$           | [1, 5]              |
| Matching function elasticity  | $\nu$                  | 0.50                |
| Ability distribution mean     | $\mu_\zeta$            | 0                   |
| Ability distribution SD       | $\sigma_\zeta$         | 1                   |
| Private cost of education     | $[Y(e_2), Y(e_3)]$     | [0.00, 0.48]        |
| Education cost function scale | $[k_0(e_2), k_0(e_3)]$ | [2.78, 4.28]        |
| Hourly minimum wage           | $w^b$                  | 0.57                |
| Family status                 | $f$                    | <b>0C 1C 2C 3C+</b> |
| EITC Phase 1 limit            | $w_m(f)$               | 0.29 0.44 0.61 0.61 |
| EITC Phase 2 limit            | $w(f)$                 | 0.36 0.80 0.80 0.80 |
| EITC Phase 3 limit            | $\bar{w}(f)$           | 0.65 1.73 1.96 2.10 |
| EITC maximum tax credit       | $c_{\max}(f)$          | 0.02 0.15 0.24 0.28 |

Notes: Family status: no children (0C), 1 child (1C), 2 children (2C), 3 children and more (3C+).

(a) Table 4: calibrated parameters

TABLE 5  
Education targets

|       | Benchmark value |        |       | $\nearrow c_{\max}$ |        |       |
|-------|-----------------|--------|-------|---------------------|--------|-------|
|       | Low             | Middle | High  | Low                 | Middle | High  |
| Data  | 9.48            | 30.69  | 59.83 | 10.18               | 29.99  | 59.83 |
| Model | 9.48            | 30.69  | 59.83 | 10.18               | 29.99  | 59.83 |

Notes: Interpretation: a 10% increase in  $c_{\max}$  increases the percentage of low-educated workers from 9.48% to 10.18%, which represents a 7.4% increase in the proportion of low-educated workers.

TABLE 6  
Estimated parameters

| Parameter                            | Symbol       | Education |      |      |
|--------------------------------------|--------------|-----------|------|------|
|                                      |              | Low       | Mid  | High |
| Curvature - human capital            | $\gamma$     | 0.41      | 0.64 | 0.60 |
| Curvature - hourly wage              | $\bar{a}$    | 0.36      | 0.60 | 0.82 |
| Matching efficiency                  | $\chi$       | 0.14      | 0.13 | 0.12 |
| Vacancy posting cost                 | $c$          | 0.28      | 0.48 | 0.54 |
| Exogenous separation rate            | $s$          | 0.06      | 0.04 | 0.03 |
| TFP                                  | $A$          | 0.89      | 0.97 | 1.26 |
| Labour disutility scale              | $\bar{q}$    | 1.12      | 1.35 | 1.34 |
| Labour disutility curvature (x10000) | $\rho$       | 0.12      | 0.19 | 0.11 |
| Education cost function growth rate  | $\phi_1(e)$  | 0         | 0.31 | 6.34 |
| Education cost function elasticity   | $\phi_2(e)$  | 0         | 1.18 | 0.27 |
| Common                               |              |           |      |      |
| Inverse of the Frisch elasticity     | $\phi$       |           |      | 1.18 |
| Prob. switch human cap.              | $\psi_n$     |           |      | 0.07 |
| Prob. switch human cap.              | $\psi_u$     |           |      | 0.52 |
| Search cost scale                    | $k_0$        |           |      | 0.56 |
| Search cost curvature                | $\zeta$      |           |      | 0.98 |
| Persistence of idio. prod.           | $\rho_\pi$   |           |      | 0.77 |
| Standard deviation of idio. prod.    | $\sigma_\pi$ |           |      | 0.25 |

(b) Tables 5–6: education targets and estimated parameters

## 4.5 Figures 8–10: Model fit for life-cycle profiles

**Read:**

- Figure 8 compares model and data for hourly wages, hours worked, employment, and monthly wages by education group.
- Figure 9 compares the model and data for earned-income deciles across the life cycle.
- Figure 10 shows the share of workers in each EITC phase over the life cycle.
- The model is designed to match the central life-cycle patterns, not every high-income tail feature.

**Economic message:** The model is credible for policy counterfactuals only if it captures life-cycle wage, hours, employment, income, and EITC-phase exposure. These figures show the fit that motivates using the model for long-run simulations.

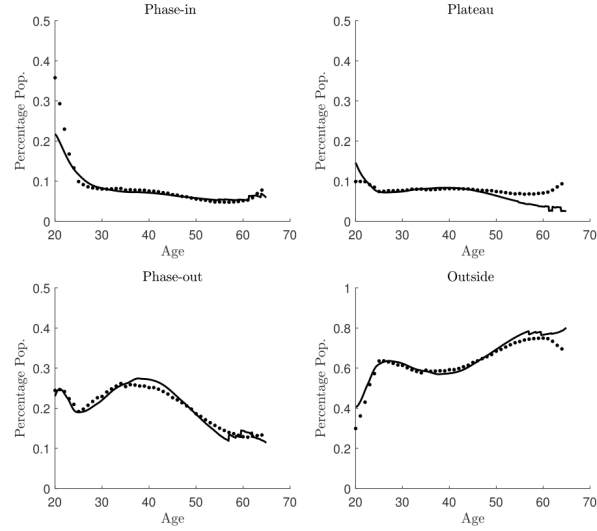


FIGURE 10  
Proportion of workers in each phase of the EITC schedule

Source: CPS, 2005–18. Dots: data, solid line: model

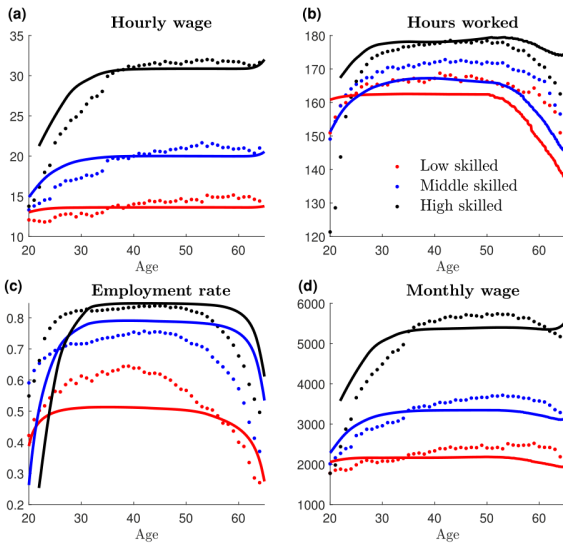


FIGURE 8  
Life-cycle profile by educational attainment

Source: CPS, 2005–18. Note: Panel (b) shows the number of hours worked per month. See Appendix H for more details on the construction of life-cycle profiles. Dots: data, solid line: model

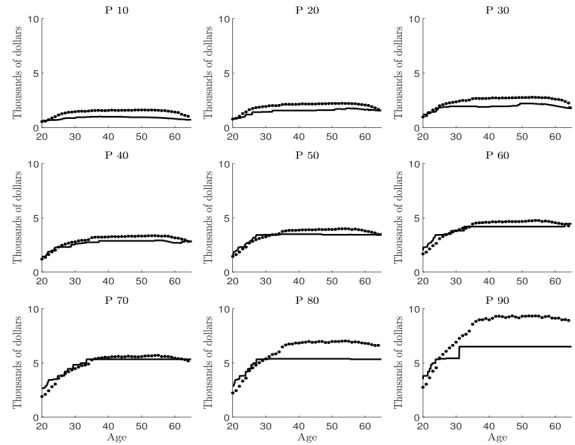


FIGURE 9  
Earned income deciles over the life cycle

Source: CPS, 2005–18. Note: Monthly earned income (hourly wage rate times hours worked) for all employed workers at each age.

(a) Figure 8: life-cycle profiles by education

(b) Figure 9: earned-income deciles

## 4.6 Figure 11, Table 7, Table 8, and Table 9: Effects of increasing EITC generosity

Read:

- Figure 11 illustrates the four types of EITC parameter changes; the main experiment increases the maximum credit.
- Table 7 shows that increasing the maximum credit changes hours and earnings differently across EITC phases.

- Table 8 shows that a 10 percent increase in maximum credit raises the low-skill share substantially and reduces the middle-skill share.
- Table 9 decomposes aggregate effects by shutting down education, matching, search, and acceptance channels.

**Economic message:** The EITC affects the economy through multiple channels. The education composition channel is especially important because it changes the long-run skill mix of the labour force.

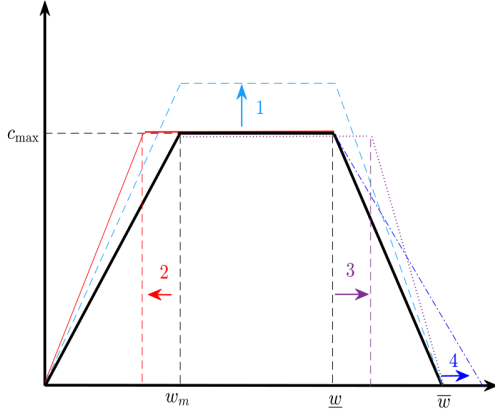


FIGURE 11  
Change in EITC

(1) 10% increase in  $c_{\max}$ , (2) 10% increase in  $w_m$ , (3) 10% increase in  $w$ , (4) 10% increase in  $\bar{w}$

(a) Figure 11: EITC parameter changes

TABLE 7  
Earning effects

| Variables     | Benchmark |         |         | $\nearrow c_{\max}$ |       |       |
|---------------|-----------|---------|---------|---------------------|-------|-------|
|               | Low       | Mid     | High    | Low                 | Mid   | High  |
| Phase-in      |           |         |         |                     |       |       |
| Hours worked  | 138.29    | 113.77  | 106.80  | 2.21                | 1.52  | 1.21  |
| Gross earning | 728.39    | 641.01  | 564.19  | 2.21                | 1.53  | 1.21  |
| Net earning   | 1008.74   | 835.54  | 701.40  | 5.09                | 4.00  | 3.311 |
| Plateau       |           |         |         |                     |       |       |
| Hours worked  | 131.35    | 129.29  | 140.35  | 0.00                | -0.00 | 0.01  |
| Gross earning | 929.13    | 1019.51 | 1145.29 | -0.04               | -0.02 | 0.05  |
| Net earning   | 1149.51   | 1321.01 | 1522.64 | 1.87                | 2.27  | 2.51  |
| Phase-out     |           |         |         |                     |       |       |
| Hours worked  | 167.79    | 150.61  | 150.61  | -0.98               | -1.41 | -1.41 |
| Gross earning | 1976.12   | 2058.47 | 1850.81 | -0.92               | -1.04 | -1.28 |
| Net earning   | 2243.31   | 2184.72 | 2085.79 | 0.56                | -0.19 | 0.22  |
| Outside EITC  |           |         |         |                     |       |       |
| Hours worked  | 176.91    | 186.66  | 186.66  | -0.01               | -0.00 | -0.00 |
| Gross earning | 2020.07   | 3298.49 | 4549.38 | -0.01               | -0.04 | -0.01 |
| Net earning   | 2020.07   | 3298.49 | 4549.38 | -0.01               | -0.04 | -0.01 |

Notes: We simulate 10,000 individuals' life-cycle trajectories and compute the average values for each EITC phase. Individuals with the same educational attainment are compared between the benchmark and the alternative scenario. We thus focus on the impact of the EITC change on hours worked and earnings, absent any change in the composition of the labour force. All values are expressed on a monthly basis. Hours worked: average number of hours worked expressed in levels in the benchmark. Earned income: hourly wage rate times hours worked. Net income: earned income + EITC (without social transfers). Earned income and net income are expressed in dollars in the benchmark. For  $\nearrow c_{\max}$ , all variable values are expressed as percentage deviations from the benchmark. Interpretation: a 10% increase in  $c_{\max}$  produces a 2.21% increase in hours worked for the low-skilled.

(b) Table 7: earnings and hours effects

TABLE 8  
Labour market effects

| Variables        | Benchmark |        |        | $\nearrow c_{\max}$ |       |       |
|------------------|-----------|--------|--------|---------------------|-------|-------|
|                  | Low       | Mid    | High   | Low                 | Mid   | High  |
| Education        | 9.48      | 30.69  | 59.83  | 7.38                | -2.28 | 0.00  |
| Hours worked     | 159.08    | 162.57 | 176.71 | -0.04               | -0.13 | -0.16 |
| Employment       | 48.44     | 73.79  | 76.08  | 0.11                | -0.03 | -0.01 |
| Search intensity | 29.91     | 41.86  | 49.04  | 0.61                | 0.12  | 0.03  |
| Contact rate     | 17.43     | 29.27  | 23.38  | -0.31               | -0.19 | -0.08 |
| Separation rate  | 12.25     | 8.65   | 5.40   | 0.00                | -0.00 | 0.00  |
| Phase-in         | 16.54     | 12.36  | 3.87   | -0.36               | -0.25 | -0.08 |
| Plateau          | 18.99     | 10.97  | 3.32   | 0.56                | 0.45  | 0.40  |
| Phase-out        | 34.20     | 27.10  | 15.48  | -0.02               | 0.08  | 0.08  |
| Outside          | 30.27     | 49.57  | 77.32  | -0.14               | -0.08 | -0.03 |

Notes: Variable values are expressed as percentage deviations from the benchmark. The results are obtained by simulating the model's steady state under the benchmark calibration and the alternative scenario by iterating on Bellman equation and stationary distribution (see [Supplementary Appendix](#)). Interpretation: a 10% increase in  $c_{\max}$  produces a 7.38% increase in the proportion of low-educated workers.

(a) Table 8: labour-market effects

TABLE 9  
Decomposing aggregate effects

| Variables        | Benchmark Level | $\nearrow c_{\max}$ |                          |                               |                                 |                                    |
|------------------|-----------------|---------------------|--------------------------|-------------------------------|---------------------------------|------------------------------------|
|                  |                 | All effects         | No $\Delta$ in education | No $\Delta$ in matching rates | No $\Delta$ in search intensity | No $\Delta$ in Acceptance decision |
| Hours worked     | 170.70          | -0.16               | -0.14                    | -0.15                         | -0.16                           | -0.16                              |
| Employment       | 72.76           | -0.25               | -0.01                    | -0.21                         | -0.29                           | -0.25                              |
| Search intensity | 45.03           | -0.09               | 0.09                     | -0.08                         | -0.20                           | -0.09                              |
| Contact rate     | 24.63           | -0.47               | -0.13                    | -0.31                         | -0.50                           | -0.47                              |
| Separation rate  | 7.05            | 0.36                | 0.00                     | 0.36                          | 0.36                            | 0.36                               |
| Prop. Phase-in   | 7.68            | 0.16                | -0.22                    | 0.12                          | 0.14                            | 0.16                               |
| Prop. Plateau    | 7.16            | 1.26                | 0.47                     | 1.21                          | 1.26                            | 1.26                               |
| Prop. Phase-out  | 20.82           | 0.30                | 0.06                     | 0.32                          | 0.28                            | 0.30                               |
| Prop. Outside    | 64.35           | -0.26               | -0.05                    | -0.25                         | -0.25                           | -0.26                              |

Notes: Variable values are expressed as percentage deviations from the benchmark. The results are obtained by simulating the model's steady state under the benchmark calibration and the alternative scenario by iterating on Bellman equation and stationary distribution (see [Supplementary Appendix](#)). The aggregate impact includes changes in the composition of workers (except in column 4) due to changes in education decisions.

(b) Table 9: aggregate decomposition

## 4.7 Figure 12: Nonlinear effects of changing the maximum credit

Read:

- Figure 12 varies the maximum credit from a large reduction to a large increase.
- The low-skill share rises as EITC generosity rises.
- Hours, earned income, net income, welfare, and government cost all respond nonlinearly.
- The redistributive gains are limited because hours and skill composition adjust.

**Economic message:** A more generous EITC is not a simple transfer. It changes the composition of the labour force and the location of workers on the EITC schedule.

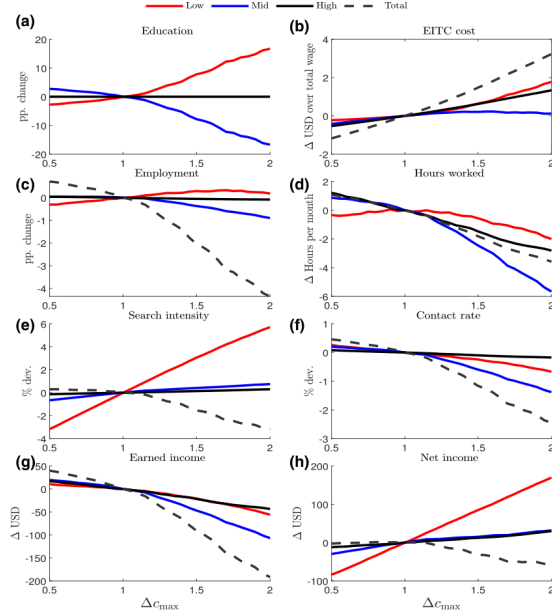


FIGURE 12  
Change in  $c_{\max}$   
 $c_{\max}$  varies between  $-50\%$  and  $+100\%$ . Earned income: hourly wage rate times hours worked. Net income: earned income + EITC (without social transfers)

#### 4.8 Table 10 and Figure 13: Optimal EITC schedules

Read:

- Table 10 compares the benchmark, no EITC, an optimal schedule without an hours constraint, and an optimal schedule with a minimum-hours constraint.
- Removing the EITC raises welfare to 101.21 relative to 100 in the benchmark, mostly by reducing education distortions.
- Optimal 1 raises welfare to 102.90 while keeping the EITC cost near the benchmark.
- Optimal 2, with a minimum-hours constraint, raises welfare to 102.98.
- Figure 13 shows that optimal schedules are shifted to the right and reach higher income ranges.

**Economic message:** The optimal EITC is broader and less concentrated at very low earnings. It supports work without making low-skill status as attractive.

TABLE 10  
Optimal tax schedule

|                  | (1)<br>Benchmark | (2)<br>No EITC | (3)<br>Optimal 1 | (4)<br>Optimal 2 |
|------------------|------------------|----------------|------------------|------------------|
| Welfare          | 100.00           | 101.21         | 102.90           | 102.98           |
| EITC cost        | 2.16             | 0.00           | 2.16             | 2.15             |
| Prop. Low edu.   | 9.48             | 6.02           | 2.64             | 2.22             |
| Prop. Mid edu.   | 30.69            | 34.15          | 37.53            | 37.95            |
| Prop. High edu.  | 59.83            | 59.83          | 59.83            | 59.83            |
| Hours worked     | 170.70           | 172.21         | 173.37           | 174.24           |
| Employment       | 72.76            | 73.61          | 74.74            | 74.90            |
| Search intensity | 45.03            | 45.04          | 45.97            | 46.03            |
| Contact rate     | 24.63            | 25.21          | 25.75            | 25.86            |
| Gross income     | 3217.36          | 3273.68        | 3313.02          | 3324.47          |
| Net income       | 3294.57          | 3273.68        | 3384.23          | 3395.52          |
| Hours constraint |                  |                |                  | 80 %             |

Notes: Welfare is defined as the discounted sum of consumption streams for every worker aged  $a$  to  $a_A$  (see Appendix K for more details). The hours constraint is expressed as a proportion of full time (maximum hours worked). Optimal 1: no constraint on hours worked. Optimal 2: constraint on hours worked. Earned income (monthly, USD): hourly wage rate times hours worked. Net income (monthly, USD): earned income + EITC (without social transfers).

(a) Table 10: optimal tax schedules

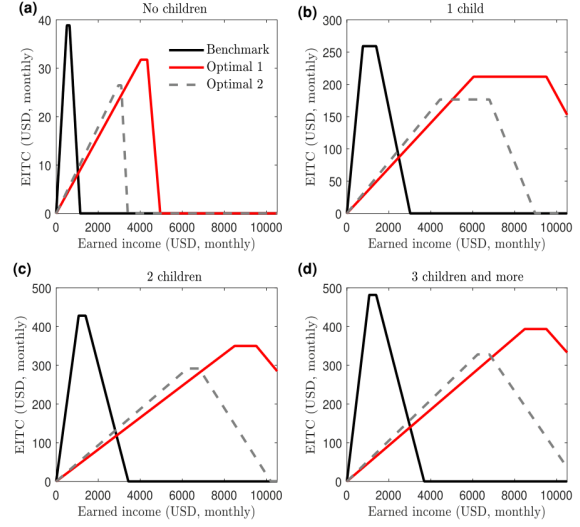


FIGURE 13  
Optimal EITC schedule - Target welfare

Note: Optimal 1: optimal EITC parameters ( $w_{in}, w_0, w, c_{max}$ ). Optimal 2: Optimal 1 + minimum hours constraint

(b) Figure 13: optimal EITC schedules

## 4.9 Figure 14: Distributional effects of alternative EITC schedules

Read:

- Figure 14 compares distributions of earned income and net income under alternative schedules.
- The EITC has little long-run impact on the overall income distribution once education and labour-market responses are included.
- Optimal schedules change incentives more than they mechanically redistribute income.

**Economic message:** The paper challenges the conventional view that in-work transfers targeted at low earnings necessarily reduce inequality in the long run.

## 4.10 Appendix robustness visuals: Table 13, Figure 15, Figure 20, and Figures 21–22

Read:

- Table 13 repeats the dropout regression with the alternative interior control group. The state EITC coefficient remains positive and significant, around 0.058.
- Figure 15 shows the ability distribution, education-cost functions, and expected value functions used to understand the education choice.
- Figure 20 displays transitional dynamics for no EITC and optimal EITC policies, showing slow composition adjustment over decades.
- Figures 21–22 show survival functions for state EITC increases. Many increases persist long enough to affect education decisions.

**Economic message:** The appendices support the main mechanism: the education response is robust, policy changes are persistent, and long-run effects require transition dynamics.



L. Transitional dynamics of optimal EITC

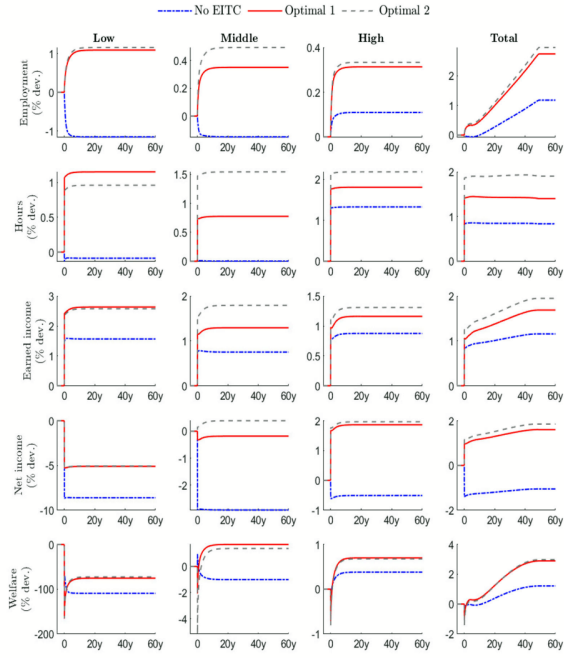


FIGURE 20  
Transitional dynamics of optimal EITC

We implement the alternative scenarios as unexpected reforms. Column (4) sums variables by skill using the proportion of individuals in each skill category, which changes as new cohorts enter the labour market

(a) Figure 20: transitional dynamics of optimal EITC

M. Survival function of an EITC increase

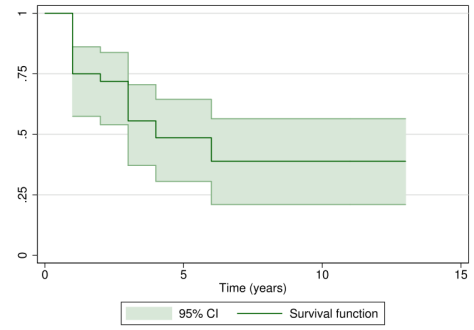


FIGURE 21  
Survival function of an EITC increase (no change in state EITC)

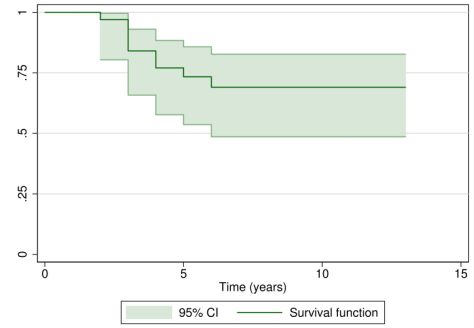


FIGURE 22  
Survival function of an EITC increase (no decrease in state EITC)

(b) Figures 21–22: survival functions of state EITC increases



## 5 Short summary

- The paper studies the EITC's long-run effects on education, labour markets, and inequality.
- The reduced-form design uses policy discontinuities at US state borders by comparing contiguous PUMA pairs.
- The ACS sample covers 2005–2018 and focuses on 18- to 24-year-old education outcomes.
- A one-percentage-point increase in the state EITC raises high-school dropout rates by about 0.07 percentage point.
- The structural model is a life-cycle directed-search model with endogenous education, human capital, search effort, hirings, hours, separations, and transfers.
- The EITC increases the value of low-skill work and therefore lowers the relative return to education.
- A more generous EITC raises the low-skill share in the long run.
- Many recipients reduce hours because they are in the phase-out or noneligible ranges.
- The long-run increase in low-skilled workers and reduction in hours limit the EITC's redistributive benefits.
- Removing the EITC can increase long-run welfare but hurts low- and middle-skilled workers in transition.
- Optimal EITC design keeps in-work incentives while reducing schooling and hours distortions.
- The welfare-improving schedules have lower maximum credits, wider phase-in and phase-out ranges, and coverage of higher earned incomes.
- A minimum-hours eligibility condition further improves performance by discouraging hours reductions.
- The broader lesson is that policy evaluation must consider pre-labour-market human-capital choices.

## 6 Conclusion

### 6.1 Core conclusion

Albertini, Poirier, and Terriau show that the EITC has important long-run effects beyond short-run labour-force participation. By subsidizing low-paid work, the EITC can reduce the incentive to complete education, increase the long-run share of low-skilled workers, reduce aggregate hours and employment, and limit redistribution.

### 6.2 Economic intuition

The EITC is intended to make work pay. But if the credit is targeted too narrowly at low earnings, it also makes low-skill work more attractive relative to schooling. Some individuals respond before entering the labour market by leaving school earlier. Employed workers may also reduce hours to qualify for the credit or increase the amount received. These responses interact with firm vacancy posting and matching, generating general equilibrium effects.

### 6.3 Policy takeaway

The paper's policy message is not simply to abolish in-work benefits. The lesson is that design matters. A broader EITC schedule that extends to higher incomes, has lower maximum benefits, and includes a minimum-hours condition can preserve participation incentives while reducing perverse effects on education and hours.